Answer Key - Health Sciences - K-12 Level Assessment 16 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) Fat = 17%, Fat Free Mass=83%, Intracellular Fluid= 40%, Extracellular Fluid= 20%; B) Fat = 83%, Fat Free Mass=40%, Intracellular Fluid= 17%, Extracellular Fluid= 20%; C) Fat = 20%, Fat Free Mass=17%, Intracellular Fluid= 40%, Extracellular Fluid= 83%; D) Fat = 40%, Fat Free Mass=20%, Intracellular Fluid= 17%, Extracellular Fluid= 83%
   Note: The correct answer reflects the typical body composition of a normal weight young adult man, where fat constitutes about 17% of total body weight, while the remaining 83% is composed of fat-free mass, which includes muscle, bone, and organs, along with the distribution of body fluids, with 40% being intracellular and 20% extracellular.

2. **Answer:** D
   Options: A) vitamin E and niacin; B) vitamin D and vitamin E; C) riboflavin and vitamin C; D) vitamin C and thiamin
   Note: Vitamin C and thiamin are water-soluble vitamins that are sensitive to heat and can easily be destroyed during cooking processes, particularly when exposed to high temperatures or prolonged cooking times.

3. **Answer:** C
   Options: A) Meat; B) Confectionary; C) Fruits and vegetables; D) Potatoes
   Note: Fruits and vegetables are often consumed in lower quantities in Wales and Scotland due to cultural dietary preferences and socioeconomic factors that affect access and availability.

4. **Answer:** A
   Options: A) Biotin; B) Niacin; C) Riboflavin; D) Thiamin
   Note: Biotin is the correct answer because it functions as a coenzyme in carboxylation reactions, facilitating the transfer of carbon dioxide in metabolic processes.

5. **Answer:** B
   Options: A) 20%; B) 40%; C) 60%; D) 80%
   Note: The correct answer of 40% reflects the significant proportion of the global population affected by untreated dental caries, highlighting a widespread oral health issue that underscores the importance of preventive care and access to dental services.

True / False

6. **Answer:** False
   Options: A) True; B) False

Note: Phytic acid primarily affects the absorption of minerals like iron, zinc, and calcium, but it does not significantly impact potassium absorption in the body.

7. **Answer:** True
   Options: A) True; B) False
   Note: Recumbent length is measured while an infant is lying down, making it an essential anthropometric measure for accurately assessing their growth and development.

8. **Answer:** False
   Options: A) True; B) False
   Note: HIV infection rates in Europe have not shown significant decreases in recent years, with many countries experiencing steady or even rising infection rates due to various factors including increased drug resistance and changes in sexual behavior.

9. **Answer:** False
   Options: A) True; B) False
   Note: Wilson's disease is a genetic disorder that leads to copper accumulation in the body, not zinc deficiency, as it affects copper metabolism rather than zinc.

10. **Answer:** True
    Options: A) True; B) False
    Note: De novo fatty acid synthesis occurs when there is an excess of energy-yielding substrates, such as carbohydrates, leading the body to convert surplus energy into fat for storage.

Long Form Response

11. **Answer:** Intrauterine growth retardation (IUGR) refers to a condition where a fetus does not grow to its expected size during pregnancy, often resulting in a birth weight that is below the population threshold for gestational age. This means that when a baby is born, its weight is lower than what is typical for its gestational age, which can indicate potential health issues. IUGR can lead to various complications for the fetus, including a higher risk of developmental delays, health problems at birth, and long-term effects on growth and learning. Understanding IUGR is crucial for monitoring fetal health and ensuring that appropriate care is provided to support the growth and development of the baby.
    Note: correct because it accurately defines intrauterine growth retardation (IUGR) as a condition where the fetus fails to achieve expected growth, resulting in a birth weight below the gestational age threshold, which has significant implications for the health and development of the newborn.

12. **Answer:** Fat-soluble vitamins, which include vitamins A, D, E, and K, play essential roles in maintaining overall health and can influence cardiovascular disease. Research suggests that these vitamins contribute to heart health by supporting functions such as maintaining healthy blood vessels and reducing inflammation. For instance, vitamin D has been linked to lower blood pressure, while vitamin E may help prevent oxidative

damage to heart tissues. Current dietary recommendations emphasize the importance of obtaining these vitamins from a balanced diet, rich in fruits, vegetables, nuts, and healthy fats, to potentially lower the risk of cardiovascular issues. Understanding the impact of fat- soluble vitamins is crucial for making informed dietary choices that promote heart health.
Note: correct because it identifies the specific fat-soluble vitamins, highlights their health benefits related to cardiovascular disease, and references current research that supports their importance in maintaining heart health and reducing disease risk.

End of Answer Key